

Manual

Energy Management - Energy Consumption Analysis EMG-VAN 8.1

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1 Energy Management - Energy Consumption Analysis

Purpose

Efficient reports monitor the development of energy consumption and complement the basic evaluation of the tabular consumption statistics.

The following functions are provided: Listing of consumption in definable periods; management of settlement periods; definition of settlement periods; printing reports.

Select a hierarchy and display a table of the company's energy counters. Show the logical energy counters assigned to machines and plant equipment and display the consumption values and comparative values. Display the counter hierarchy.

You use the function package if:

- you want to analyze consumption over time.
- you want to list specific periods in summary.
- you want to show or print data since the last settlement.
- you want to reset meter readings after the current settlement and thus limit the analyzed period.
- you want to have an overview of the current status of energy consumption since the last settlement.
- you want to have an overview of the current status of energy consumption since the last settlement.

Integration

The reports require the function package EMG-MGM as a basis for data collection.

Features

Consumption analysis

- Graphic consumption statistics: Show the values of a statistical table as bar chart.
- Show reports on the development of energy consumption in table form. Display the energy consumption of the real and the logical (calculated) meters over time. Flexible filter settings and compression functions are provided.
- Expand the monitoring of hierarchical energy data in order to evaluate historical data of definable periods.
- Compress data in manifold ways by machines, counters, report groups and other logistic objects.

Consumption statement

- List consumption in periods of your choice. List consumption in defined settlement periods. Select by logistic objects, meters, machines or machine report groups.
- Integrate KPIs like costs in the lists (using EMG-KPI).
- Report function: Create settlement overviews.
- Complete a settlement period. Manage and store settlement periods including the status.

Energy Monitor

- Display the current energy counters and power meters in a dynamic tree-like structure.
- Display machines and other logistic objects in the structure of the counters and their connections.
- Create dynamic object hierarchies considering multiple selection criteria like counters, machines, report groups and other logistic objects.
- Display counters, counters' consumption, limits and violations of limits as well as implausible configurations and values in the graphic nodes.
- Energy balance: Show the energy differences in the counter hierarchy.
- Drill-down-mechanism in the branches of a tree structure.

2 Consumption analysis

Overview

Menu	Production Facility Management → Resource analysis → Consumption analysis
Transaction code	cona
Function authorization	cona

This document describes the application "Consumption analysis" in the Manufacturing Operation Center (MOC).

Purpose

Using the consumption analysis, you can display the consumptions of selected counter resources in chronological order. The consecutive time intervals are called compressions. The compression interval can be set to hours, days, weeks and months. The listed table rows show the resource's consumption values in the corresponding compression interval as well as the values of the comparison resources. In the pivot table, you can correlate the compressed values in many ways.

When selecting values, you have the possibility to filter the resources in multiple ways in tree-like form using the defined counter hierarchies.

Integration

This application is connected with the following applications:

- Consumption monitor to display the current counter readings since the reset.
- Consumption statement for an analysis of the totals and to reset the data.

Selection criteria

Reference

Selection of resources using the resource list

Resource type / Resource / Name / Resource family

Selection criteria relating to the resource.

Period from / until

Time period considered in an evaluation

Compress preselection

Select time interval for the compression (every 15 minutes, hour, day, week, month, year):

- Select between 15 minutes and one week (EMG 8.2)
- Select between one hour and one week
- Select between one day and three months
- Select between one week and two years

Field descriptions**Resource**

Resource master data

Date

Time reference for the consumption values. The compression depends on the preselected compression and the considered time period.

Consumption

Consumption value with unit.

The accuracy indicates the quality data. The data quality is the sum total of all documents (degree of overlapping in percent * duration in the interval) / sum total (durations in the interval) .

A value 1.0 = Good and a value 0.0 = Poor.

Comparison values 1 / 2

Consumption values of the comparison resource. In addition, the deviation in absolute values and as a percentage is shown.

Detail application pivot consumption analysis

In the pivot table, you can compress even more the data by time period/counters. The corresponding bar chart illustrates the displayed values.



The bar chart of the pivot grid shows a maximum of 10 bars (for each value).

3 Consumption/Energy Monitor

Overview

Menu	Production Facility Management → Resource analysis → Energy monitor
Transaction code	conmon
Function authorization	conmon

This document provides a description of the "Consumption/ energy monitor" application in the Manufacturing Operation Center (MOC).

Purpose

The energy monitor lists the energy counters (resources). The energy monitor shows the current status for these resources, allowing a comparison with comparative resources. This makes it possible to display an energy balance. The system always shows the counted quantities since the last inventory, i.e. since the counter was reset.

You can also display the balance in a graphic. The same applies to the counter hierarchy defined by the BOM including the meter readings of the selected resource.

When you select data, you can use the defined counter hierarchies in tree-like structure to filter the resources in the table multiple times.

Integration

This application is connected with the following applications:

- Consumption analysis to chronologically distribute consumption values
- Consumption statement to analyze the totals and to reset data.
- You can also visualize the monitor values in the graphic machinery due to the integration of the counters (resources).

Selection criteria

Resource type

Type of resource.

Workplaces and machines always have the resource type MNR. But you can configure individual resource types for the other resources. Predefined resource types include:

DNC NC/DNC program

DOC	Document
ENT	Removal device
MNR	Workplace/Machine
PAC	Packaging, transportation container
PRM	Test and measuring equipment
PER	Production staff / general
PRU	Setup staff
TEM	Tempering equipment
VOR	Device
WNR	Tool

Resource from ... to ...

This selection criterion refers to the resource. You can also run a search using wildcards (placeholders * and ?).

Designation (name)

Name of the resource.

Cost center

Cost center of the resource.

Responsibility area

Responsibility area to which the resource is assigned.

Resource family

The resource family to which the resource is assigned.

Storage location

Regular storage location of the resource.

User fields

MD user fields 1- 6 of the resource.

Reference

The resource's internal ID.

Field descriptions*"General" category***Resource type**

Resource type of the resource. By default, the HYDRA system includes some default resource types. You can configure additional resource types in HYDRA.

Resource

Resource as a reference point for comparisons.

Designation (name)

Name of the resource.

Resource status

Resource status.

Cost center

Cost center

*"Resource" category***Current value**

Current value

Quantity unit (P)

Quantity unit (P)

Absolute value limit (only available if the EMG-KBW license has been purchased)

Absolute value limit

Reset time

Date on which the resource counters were reset.

Status time

Status time

"Key figures" category (only available if the EMG-KBW license has been purchased)**Energy consumption per hour**

Energy consumption per hour

Key figure 2 to 10

You can show up to nine additional key figures. Use the formula management to define the key figures (KPIs). You can configure the formulas eovf2 to eovf10.

*"Comparison resource 1/2" category***Resource 1 / Resource 2**

Resource whose values are to be compared with those of the reference point.

Current value

Absolute difference between the values of the reference resource and the comparison resource.

Difference, in percent

Relative difference between the reference resource values and the comparison resource values.

Unit

Unit

Resource type 1 / Resource type 2

Resource type of the comparison resource.

Reset time

Date on which the counters of the comparison resource were reset.

Status time

Status time

Setting a reset time

The reset date depends on the entered records.

If you select a reset date that coincides with a resource record period, the system sets the reset time to the start time of this resource record.



If you select a reset date that does not coincide with a resource record period, the system uses the start time of the next resource record as the reset time.

The system uses the reset time you enter if:

- you select a reset date that does not coincide with a resource record period and
- there is no record that starts at a later point in time.

4 Consumption Statement

Overview

Menu	Operating facilities management→Production facility analysis →Consumption statement
Transaction code	constat
Function authorization	constat

This document provides a description of the "Consumption statement" application in the Manufacturing Operation Center (MOC).

Usage

The consumption statement lists each of the energy counter resources. It shows the current counter reading for these resources since the last reset. You can create a report that shows this data. The counter readings can be reset in this period. To do so, select the resources using the multiple selection option in the table. Furthermore, you can also enter a random period of time for the report so that not only the flexible areas since the last statement are visible, but there is also the option to select random periods and view each of the counter totals in these periods.

There is the option for the selection to achieve multiple filter levels in tree-like fashion by filtering the resources in the table using the defined counter hierarchy.

Integration

This application is connected with the following applications

- Consumption analysis to chronologically distribute consumptions and to comparatively analyze them.
- Energy/ consumption monitor to analyze the counter reading and the comparison resources.
- The monitor values can also be visualized in the graphic park by integrating the counter resources.

Selection criteria

Reference

Resources selection via the resource list

Resource type/ resource/ designation/ resource family

Selection criteria relating to resource

Last statement to

This options is initially preselected. The start date equals the last reset time of the resource in each case. The end date is predefined as "yesterday".

Thus, the report evaluates all documents that have an end time greater than the last reset time of the resource and that have an end time smaller or equal to the end date entered.

As such, the report shows the consumption value in this period, the start date of the period and the end date of the last document.

A reset can only be performed with this selection.

Period from/ to

Random time period considered in the evaluation.

A reset is not possible with this selection.

Field descriptions**Resource category**

Resource master data

Energy consumption category

Consumption value with unit, specifying the period under consideration as well, i.e. from the start of the first document to the end of the last document.

Toolbar**Reset**

Function with which to reset a statement.

For all of the selection resources (multiple selection possible), the reset date is set to the last date value for the documents in the period and the counter in the status is set to zero. (So, the reset date always shows the end stamp of the last reading on this day as the time stamp. The remaining surplus for the day will be included in the document for the next day and is settled the next time.

Resetting is only possible if it is possible for the selected resource.

**Consumption statement**

Opens the "Consumption statement" report.